

obtained: m.p. 242°C; IR (KBr) 3220 (NH), 2210 (CN), 1645 (CO); ^1H NMR (DMSO- d_6) δ (p.p.m.) 2.03 (q, J 7.3 Hz, 2H, CH_2), 2.64 (t, J 7.3 Hz, 2H, CH_2), 2.80 (t, J 7.3 Hz, 2H, CH_2), 8.00 (s, 1H, H4), 12.72 (s, 1H, NH).

2-Chloro-5-ethyl-6-methylpyridine-3-carbonitrile **4a**

A solution of **3a** (48.7 g, 0.3 mol), phosphorus pentachloride (68.7 g, 0.33 mol) in phosphorus oxychloride (200 mL) was heated to 80°C for 4 h. Phosphorus oxychloride was evaporated under reduced pressure. The resulting oil was poured cautiously into ice-water and the semi-solid was extracted with chloroform (3×100 mL). The organic extract was washed successively with saturated aqueous sodium hydrogenocarbonate, water and brine. The organic layer was dried over magnesium sulphate and concentrated under reduced pressure to yield 40.5 g (75%) of a pale yellow solid: m.p. 64°C; IR (KBr) 2220 (CN); ^1H NMR (CDCl_3) δ (p.p.m.) 1.27 (t, J 7.3 Hz, 3H, CH_3), 2.59 (s, 3H, CH_3), 2.67 (q, J 7.3 Hz, 2H, CH_2), 7.73 (s, 1H, H4).

2-Chlorocyclopenta[b]pyridine-3-carbonitrile **4b**

This compound was prepared similarly to **4a** starting from **3b** (48.1 g, 0.3 mol). 38.1 g (71%) of a pale yellow solid was obtained: m.p. 130°C; IR (KBr) 2220 (CN); ^1H NMR (CDCl_3) δ (p.p.m.) 2.23 (q, J 7.3 Hz, 2H, CH_2), 3.00 (t, J 7.3 Hz, 2H, CH_2), 3.09 (t, J 7.3 Hz, 2H, CH_2), 7.77 (s, 1H, H4).

5-Ethyl-2-methoxy-6-methylpyridine-3-carbonitrile **5a**

A stirred solution of sodium methoxide (11.9 g, 0.22 mol) in methanol (200 mL) was cooled to 0°C. 2-chloropyridine-3-carbonitrile **4a** (36.1 g, 0.2 mol) was then added and the resulting mixture refluxed for 2 h. Methanol was evaporated under reduced pressure and the resulting solid poured into water (100 mL) and extracted with diethylether (300 mL). The organic layer was washed successively with a 1 M solution of hydrochloric acid (100 mL), saturated aqueous sodium hydrogenocarbonate (100 mL) then water (2×100 mL), dried over magnesium sulphate, filtered and the solvent was evaporated under reduced pressure to give a white solid (32.9 g, 93%): m.p. 62°C; IR (KBr) 2220 (CN); ^1H NMR (CDCl_3) δ (p.p.m.) 1.19 (t, J 7.3 Hz, 3H, CH_3), 2.48 (s, 3H, CH_3), 2.57 (q, J 7.3 Hz, 2H, CH_2), 4.01 (s, 3H, CH_3O), 7.59 (s, 1H, H4).

2-Methoxycyclopenta[b]pyridine-3-carbonitrile **5b**

This compound was prepared similarly to **5a** starting from **4b** (32.1 g, 0.18 mol). 26.3 g (84%) of a white solid was obtained: m.p. 115°C; IR (KBr) 2220 (CN); ^1H NMR

(CDCl_3) δ (p.p.m.) 2.16 (q, J 7.3 Hz, 2H, CH_2), 2.88 (t, J 7.3 Hz, 2H, CH_2), 2.97 (t, J 7.3 Hz, 2H, CH_2), 4.03 (s, 3H, CH_3O), 7.64 (s, 1H, H4); ^{13}C NMR (CDCl_3) δ (p.p.m.) 23.1 (CH_2), 29.5 (CH_2), 34.8 (CH_2), 54.3 (CH_3O), 93.3 (Pyr), 116.2 (CN), 129.3–138.0–164.3 and 169.0 (Pyr). Anal. calcd for $\text{C}_{10}\text{H}_{10}\text{N}_2\text{O}$: C, 68.95; H, 5.79; N, 16.08. Found: C, 68.54; H, 5.84; N, 15.87.

5-Ethyl-2-methoxy-6-methylpyridine-3-carboxaldehyde **6a**

A 1 M solution of diisobutylaluminium hydride (Dibal-H) in tetrahydrofuran (THF) (120 mL, 0.12 mol) under an argon atmosphere was cooled to 0°C. A solution of **5a** (17.6 g, 0.1 mol) in anhydrous THF (50 mL) was added over a period of 15 min. The reaction mixture was stirred at room temperature overnight. The solvent was evaporated under reduced pressure and the resulting oil was cautiously poured into a mixture of ice-cooled 1 M hydrochloric acid (150 mL) and diethylether (200 mL). After 1 h the organic layer was separated, washed with brine (22×100 mL), dried over magnesium sulphate, filtered through a pad of charcoal and concentrated. Trituration of the residual oil gave a white solid (10.4 g, 58%): m.p. 56°C; IR (KBr) 1670 (CO); ^1H NMR (CDCl_3) δ (p.p.m.) 1.21 (t, J 7.3 Hz, 3H, CH_3), 2.49 (s, 3H, CH_3), 2.59 (q, J 7.3 Hz, 2H, CH_2), 4.03 (s, 3H, CH_3O), 7.85 (s, 1H, H4), 10.31 (s, 1H, CHO); ^{13}C NMR (CDCl_3) δ (p.p.m.) 14.0 (CH_3), 22.4 (CH_3), 24.6 (CH_2), 53.5 (CH_3O), 116.5–130.4–136.8–161.5 and 162.2 (Pyr), 189.3 (CHO).

2-Methoxycyclopenta[b]pyridine-3-carboxaldehyde **6b**

This compound was prepared similarly to **6a** starting from **5b** (20.9 g, 0.12 mol). 11.8 g (55%) of a white solid was obtained: m.p. 75°C; IR (KBr) 1660 (CO); ^1H NMR (CDCl_3) δ (p.p.m.) 2.14 (q, J 7.3 Hz, 2H, CH_2), 2.85 (t, J 7.3 Hz, 2H, CH_2), 2.92 (t, J 7.3 Hz, 2H, CH_2), 4.03 (s, 3H, CH_3O), 7.90 (s, 1H, H4), 10.31 (s, 1H, CHO); ^{13}C NMR (CDCl_3) δ (p.p.m.) 23.0 (CH_2), 29.5 (CH_2), 34.8 (CH_2), 53.8 (CH_3O), 116.4–130.0–132.8–164.7 and 170.7 (Pyr), 189.2 (CHO). Anal. calcd for $\text{C}_{10}\text{H}_{11}\text{NO}_2$: C, 67.78; H, 6.26; N, 7.90. Found: C, 67.51; H, 5.98; N, 7.95.

5-Ethyl-3-hydroxymethyl-2-methoxy-6-methylpyridine **7a**

A stirred solution of **6a** (9 g, 0.05 mol) in methanol (100 mL) was cooled to 0°C. Sodium borohydride (2.1 g, 0.055 mol) was added in portions and the reaction mixture was stirred for 4 h at room temperature. The solvent was evaporated under reduced pressure and the residue poured into water (100 mL) and acidified with a 1 M aqueous hydrochloric acid solution. The semi-solid was extracted